1. Which of the following are greater than x when $x = \frac{9}{11}$? **(Simplification) NIMCET-2008**

I. $\frac{1}{x}$ II. $\frac{x+1}{x}$ III. $\frac{x+1}{x-1}$

(a)I only (b)I and II only

(c)I and III only (d)II and III only

2. Let x, y and z be distinct integers, x and y are odd and positive, and z is even and positive. Which one of the following statements cannot true? **(Simplification) NIMCET-2008**

(a)$(x-z)^2$ y is even (b)$(x-z)\,y^2$ is odd

(c)$(x-z)$ y is odd (d)$(x-y)^2$ z is even

3. Correct the following equation by inter-changing two signs. 3 – 9 × 27 + 9 ÷ 3 = 3 **(Simplification) NIMCET-2011**

(a)+ and – (b)× and +

(c)× and ÷ (d)× and –

ANSWER

1	b	2	a	3	d				

SOLUTION

1. (b)

At $x = \frac{9}{11}$; $\frac{1}{x} = \frac{11}{9} > x$:

At $x = \frac{9}{11}$; $\frac{x+1}{x} = \frac{20}{9} > x$

At $x = \frac{9}{11}$; $\frac{x+1}{x-1} = 10 < x$

Hence, I and II are correct.

2. (a)

Odd-even = Odd, so $(x - z)^2$ y will be odd. Hence, (a) cannot be true.

3. (d) Given equation, 3 – 9 × 27 + 9 ÷ 3 = 3

From option (d), interchanging the signs

3 × 9 – 27 + 9 + 3 = 3

⟹ 27 – 27 + 3 = 3

⟹ 3 = 3